Section 5.2 Properties of the Determinant

The determinant can tell us a lot, such as weather a system has a unique solution or not, or weather a set of vectors is linearly independent. As such it is nice know what happens to it when we multiply by other matrices or perform row operations, or take an inverse.

Let A be a square matrix

(a) *Suppose that B is produced by interchanging rows of A. Then* $\det(A) = -\det(B)$

Let's just compare the cofactor expansions across R_i of A and R_{i+1} of B

(b) *Suppose that B is produced by multiplying a row of A by c. Then* $\det(A) = \frac{1}{c}\det(B)$

Let's look at the cofactor expansion across R_i which was multiplied by c

(c) *Suppose that B is produced by adding a multiple of one row of A to another. Then* $\det(A)$

Let's just compare the cofactor expansions across $\widehat{R_j}$ of B wich is equal to $cR_i + R_j$ of A remembering that the submatrices will be the same across both rows.

Section 5.2 Properties of the Determinant

Why is $a_{i1}C_{j1} + a_{i2}C_{j2} + \cdots a_{in}C_{jn} = 0$ if $j \neq i$ A 3×3 example

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Let $i = 2, j = 1$ then our determinant is

$$a_{21}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{22}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{23}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$a_{21}(a_{22}a_{33} - a_{32}a_{23}) - a_{22}(a_{21}a_{33} - a_{31}a_{23}) + a_{23}(a_{21}a_{32} - a_{31}a_{22})$$

$$a_{21}a_{22}a_{33} - a_{21}a_{32}a_{23} - a_{22}a_{21}a_{33} + a_{22}a_{31}a_{23} + a_{23}a_{21}a_{32} - a_{23}a_{31}a_{22} = 0$$

i could have been anything except 1! *(to get zero that is)*

$$a_{i1}a_{22}a_{33} - a_{i1}a_{32}a_{23} - a_{i2}a_{21}a_{33} + a_{i2}a_{31}a_{23} + a_{i3}a_{21}a_{32} - a_{i3}a_{31}a_{22} = 0$$

In fact we could show this with general i and each $j \neq i$ to show all cases here.

This fact is general for all matrices! Kind of cool.

Let A be a square matrix

(a) Suppose that B is produced by interchanging rows of A. Then $\det(A) = -\det(B)$

(b) Suppose that B is produced by multiplying a row of A by c. Then $\det(A) = \frac{1}{c}\det(B)$

(c) Suppose that B is produced by adding a multiple of one row of A to another. Then $\det(A)$

Section 5.2 Properties of the Determinant

First Let's Prove that $\det(AB) = \det(A)\det(B)$, to do so we need the following lemma

If both E, B are $n \times n$ matrices and E is an elementary matrix then $\det(EB) = \det(E)\det(B)$

Let E_1 be an elementary matrix for row swap.

Let E_2 be an elementary matrix for multiplication of a row.

Let E_3 be an elementary matrix for $cR_i + R_j$.

$\det(AB) = \det(A)\det(B)$

Let $E_1, E_2, \ldots E_k$ be the elementary rows which take A to rref form that is $E_k \cdots E_2 E_1 A = rref(A)$

Case 1 $\det(A) = 0$ meaning A in rref form has a zero in a pivot position.

Case 2 $\det(A) \neq 0$ in which case A has a 1 in each pivot position, which means $E_k \cdots E_2 E_1 A = I$ or that $A = (E_k \cdots E_2 E_1)^{-1} I = (E_k \cdots E_2 E_1)^{-1} = E_1^{-1} E_2^{-1} \cdots E_k^{-1}$.

Section 5.2 Properties of the Determinant

If A is a square invertible matrix

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

Proof

Consider $\det(AA^{-1})$

Let A be a square matrix, A is invertable if and only if $\det(A) \neq 0$

Suppose A has an inverse, then rref(A)=I how do $\det(A)$ and $\det(I)$ compare?

Suppose $\det(A) \neq 0$ and consider the matrix B, the echelon form of A, obtained only through Row swaps and additive row operations. What must the structure of B be?

Section 5.2 Properties of the Determinant

Let A be a square matrix

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Find the determinant of the original matrix A which was transformed to B

$$\begin{aligned} R_2 &\leftrightarrow R_3 \\ R_2 + R_3 &\rightarrow R_3 \\ -7R_2 + R_4 &\rightarrow R_4 \\ R_3 &\leftrightarrow R_4 \\ -5R_4 &\rightarrow R_4 \end{aligned} \quad \Rightarrow \quad B = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 3 & 1 & 1 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$